



CSE-476: Introduction to Natural Language Processing
Quiz 1

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

- Q1. In the BIO tagging scheme for NER, what does the ``B-'' prefix indicate?
- (A) Between two tokens
 - (B) Beginning of an entity
 - (C) Inside an entity
 - (D) Outside any entity
- Q2. Which statement best describes a (plain) Markov chain in sequence tagging?
- (A) It uses only words (observations) and ignores labels
 - (B) It models transitions between labels but does not consider the words
 - (C) It models emissions $P(\text{word} \mid \text{label})$
 - (D) It is identical to an HMM
- Q3. In a Hidden Markov Model (HMM) for NER, an emission probability is:
- (A) $P(\text{label} \mid \text{previous label})$
 - (B) $P(\text{word} \mid \text{label})$
 - (C) $P(\text{label} \mid \text{word})$
 - (D) $P(\text{word} \mid \text{previous word})$
- Q4. In an n-gram language model, a bigram LM predicts the next word using:
- (A) No previous words
 - (B) The previous 1 word
 - (C) The previous 2 words
 - (D) The entire sentence history
- Q5. Which statement about perplexity is correct?
- (A) It's the raw probability of a sentence

- (B) It's the inverse probability of a test text, normalized by length
 - (C) It's the fraction of correctly labeled tokens
 - (D) It's the number of unique words in the corpus
- Q6. Which of the following best describes the main goal of tokenization in NLP?
- (A) To label each token with a grammatical category
 - (B) To convert text into a sequence of symbols suitable for modeling
 - (C) To reduce vocabulary size to zero
 - (D) To assign probabilities to sentences
- Q7. Why is splitting text by spaces (`text.split(" ")`) considered an insufficient tokenizer?
- (A) It produces too many tokens
 - (B) It does not work well for languages without explicit word boundaries
 - (C) It cannot tokenize English at all
 - (D) It ignores punctuation entirely
- Q8. Which of the following is NOT listed as a common sequence tagging task?
- (A) Part-of-speech tagging
 - (B) Named Entity Recognition
 - (C) Dependency parsing
 - (D) Sentiment analysis
- Q9. What is a common limitation of BIO tagging for Named Entity Recognition?
- (A) It cannot represent overlapping or nested entities
 - (B) It requires too many labels
 - (C) It does not support multiple entity types
 - (D) It ignores entity boundaries
- Q10. When choosing between character-level tokenization and sentence-level tokenization, which trade-off must be considered?
- (A) Speed vs accuracy
 - (B) Tokenized sequence length vs vocabulary size
 - (C) Precision vs recall
 - (D) Memory vs computation

End of Paper